$$V = V_{int} + V_{bool} + V_{fun} + V_{ref} + V_{tuple}$$
$$D = (\{\textbf{error}, \textbf{typeerror}\} + \Sigma \times V)_\perp$$
$$Env = \langle var \rangle \to V$$

La función ι_{norm} aceptará ahora un par estado-valor: $\iota_{norm}\langle \sigma, z\rangle \in D$.

$$
\begin{aligned}
f_*(\iota_{norm}\langle \sigma, z\rangle) &= f(\langle \sigma, z\rangle)\\
f_*(err) &= err\\
f_*(tyerr) &= tyerr\\
f_*(\perp) &= \perp
\end{aligned}
$$

$$
\begin{aligned}
f_{int}(\langle \sigma, \iota_{int}\, k\rangle) &= f(\langle \sigma, k\rangle)\\
f_{int}(\langle \sigma, \iota_{\theta}\, z\rangle) &= tyerr \qquad \text{si } \theta \neq int
\end{aligned}
$$

La semántica de los construcciones típicamente imperativas es la siguiente:

$$[\![\textbf{val}\ e]\!]\eta\sigma = \left(\lambda\langle\sigma', r\rangle.\ \begin{cases}\iota_{norm}\langle\sigma', \sigma' r\rangle & \text{si } r \in dom(\sigma')\\ err & \text{si no}\end{cases}\right)_{ref*}([\![e]\!]\eta\sigma)$$

$$[\![\textbf{ref}\ e]\!]\eta\sigma = (\lambda\langle\sigma', z\rangle.\ \iota_{norm}\langle[\sigma' \mid r_{\sigma'} : z], \iota_{ref}\, r_{\sigma'}\rangle)_*([\![e]\!]\eta\sigma)$$
$$\text{donde}\ r_{\sigma'} = new(\sigma')$$

$$[\![e := e']\!]\eta\sigma = (\lambda\langle\sigma', r\rangle.\ (\lambda\langle\hat\sigma, z\rangle.\ \iota_{norm}\langle[\hat\sigma \mid r : z], \langle\rangle_V\rangle)_*[\![e']\!]\eta\sigma'))_{ref*}([\![e]\!]\eta\sigma)$$

$$[\![e =_{ref} e']\!]\eta\sigma = (\lambda\langle\sigma', r\rangle.\ (\lambda\langle\hat\sigma, r'\rangle.\ \iota_{norm}\langle\hat\sigma, \iota_{bool}\, r = r'\rangle)_{ref*}$$
$$([\![e']\!]\eta\sigma'))_{ref*}([\![e]\!]\eta\sigma)$$

$$[\![0]\!]\eta\sigma = \iota_{norm}\langle\sigma, \iota_{int}\, 0\rangle$$
$$[\![\textbf{true}]\!]\eta\sigma = \iota_{norm}\langle\sigma, \iota_{bool}\, T\rangle$$

$$[\![-e]\!]\eta\sigma = (\lambda\langle\sigma', i\rangle.\ \iota_{norm}\langle\sigma', \iota_{int}\, -i\rangle)_{int*}([\![e]\!]\eta\sigma)$$

$$[\![e + e']\!]\eta\sigma = (\lambda\langle\sigma', i\rangle$$
$$(\lambda\langle\hat\sigma, j\rangle.\ \iota_{norm}\langle\hat\sigma, \iota_{int}\, i + j\rangle)_{int*}([\![e']\!]\eta\sigma'))_{int*}([\![e]\!]\eta\sigma)$$

peradores típicos del cálculo lambda se adaptan como sigue:

$$
\begin{aligned}
[\![v]\!]\eta\sigma &= \iota_{norm}\langle\sigma, \eta\ v\rangle\\
[\![e\, e']\!]\eta\sigma &= (\lambda\langle\sigma', f\rangle.\ f_*([\![e']\!]\eta\sigma'))_{fun*}([\![e]\!]\eta\sigma)\\
[\![\lambda v.\ e]\!]\eta\sigma &= \iota_{norm}\langle\sigma, \iota_{fun}(\lambda\langle\sigma', z\rangle.\ [\![e]\!][\eta \mid v : z]\sigma')\rangle
\end{aligned}
$$

$$[\![\textbf{letrec}\ w = \lambda v.e\ \textbf{in}\ e']\!]\eta\sigma = [\![e']\!][\eta \mid w : \iota_{fun}\, g]\sigma$$

donde

$$g : V_{fun} \qquad\qquad F : V_{fun} \to V_{fun}$$
$$g = \mathbf{Y}_{V_{fun}} F \qquad\qquad F(f)(\langle\sigma', z\rangle) = [\![e]\!][\eta | w : \iota_{fun}\, f | v : z]\sigma'$$

3. Demuestre la propiedad que describe la semántica denotacional de la declaración de variables locales, donde $r_{\sigma'} = new(dom\ \sigma')$.

$$[\![\mathbf{newvar}\ v = e\ \mathbf{in}\ e']\!]\eta\sigma = (\lambda\,\langle\sigma', z\rangle\,.\ [\![e']\!][\eta|v : \iota_{ref} r_{\sigma'}][\sigma'|r_{\sigma'} : z])_*([\![e]\!]\eta\sigma)$$

4. Para el lenguaje aplicativo eager con referencias y asignación, calcular la semántica de denotacional de $x := 1 + \mathbf{val}\ x$ en:

   a. η, σ tales que $\eta x = \iota_{ref} r$, $\sigma r = \iota_{int} 1$

   b. η, σ tales que $\eta x = \iota_{ref} r$, $\sigma r = \iota_{bool}\mathbf{True}$

   c. η, σ tales que $\eta x = \iota_{int} 0$

$$[\![e := e']\!]\eta\sigma = (\lambda\langle\sigma', r\rangle.\ (\lambda\langle\hat\sigma, z\rangle.\ \iota_{norm}\langle[\hat\sigma\,|\,r : z], \langle\rangle_V\rangle)_*[\![e']\!]\eta\sigma')_{ref*}([\![e]\!]\eta\sigma)$$

a) $[\![x := 1 + \mathrm{val}\ x]\!] \overbrace{[\eta | x : \iota_{ref}\ r]}^{\eta_0}\ \overbrace{[\sigma | r : \iota_{int}\ 1]}^{\sigma_0} =$

$= (\lambda\langle\sigma', r\rangle.\ (\lambda\langle\hat\sigma, z\rangle.\ \iota_{norm}\langle[\hat\sigma | r : z], \langle\rangle_V\rangle)_* [\![1 + \mathrm{val}\ x]\!]\eta_0\sigma'))_{ref*}([\![x]\!]\eta_0\sigma_0)$

$= (\quad (\quad ...))_{ref*}\ \iota_{norm}\langle\sigma_0, \iota_{ref}\ r\rangle$

$= \quad))\ \langle\sigma_0, r\rangle$

σ' $\quad r$

$= (\lambda\langle\hat\sigma, z\rangle.\ \iota_{norm}\langle[\hat\sigma | r : z], \langle\rangle_V\rangle)_* [\![1 + \mathrm{val}\ x]\!]\eta_0\ \sigma_0$

$[\![1 + \mathrm{val}\ x]\!]\eta_0\sigma_0 = (\lambda\langle\sigma', i\rangle\ (\lambda\langle\hat\sigma, j\rangle.\ \iota_{norm}\langle\hat\sigma, \iota_{int}(i+j)\rangle)_{int*}([\![1]\!]\eta_0\sigma'))_{int*}([\![\mathrm{val}\ x]\!]\eta_0\sigma_0)$

$[\![\mathrm{val}\ x]\!]\eta_0\sigma_0 = \left(\lambda\langle\sigma', r\rangle.\ \begin{cases}\iota_{norm}\langle\sigma', \sigma' r\rangle & \text{si } r \in dom(\sigma') \\ err & \text{c.c.}\end{cases}\right)_{ref*} [\![x]\!]\eta_0\sigma_0$

$= (\quad \cdot \quad - \quad - \quad)_{ref*}\ \iota_{norm}\langle\sigma_0, \iota_{ref}\ r\rangle$

$= \left(\lambda\langle\sigma', r\rangle \begin{cases}\iota_{norm}\langle\sigma', \sigma' r\rangle & \text{si } \sim \\ \sim & -\end{cases}\right) r\ \langle\sigma_0, r\rangle$

σ' $\quad r$

$$= \begin{cases} \iota_{norm} \langle \sigma_0, \sigma_0 t \rangle & \text{si } t \in \text{Dom}(\sigma_0) \\ \text{c/n} & \text{s.c.} \end{cases}$$

$\sigma_0 t = \iota_{int}\, 1$
¿existe?

$$= \iota_{norm} \langle \sigma_0, \iota_{int}\, 1 \rangle$$

$$[\![1 + \text{val } x]\!] \eta_0\, \sigma_0 = (\lambda \langle \sigma', i \rangle\ (\lambda \langle \hat{\sigma}, j \rangle . \iota_{norm} \langle \hat{\sigma}, \iota_{int}(i+j) \rangle)_{int *} ([\![1]\!] \eta_0 \sigma'))_{int *} ([\![\text{val } x]\!] \eta_0 \sigma_0)$$

$$= (\lambda \langle \sigma', i \rangle\ (\lambda \langle \hat{\sigma}, j \rangle . \iota_{norm} \langle \hat{\sigma}, \iota_{int}(i+j) \rangle)_{int *} ([\![1]\!] \eta \sigma'))_{int *}\ \iota_{norm} \langle \sigma_0, \iota_{int}\, 1 \rangle$$

$$= (\qquad\qquad)\qquad(\qquad))\ \langle \sigma_0, 1 \rangle \rightarrow \langle \sigma', i \rangle$$

$$= (\lambda \langle \hat{\sigma}, j \rangle . \iota_{norm} \langle \hat{\sigma}, \iota_{int}(1+j) \rangle)_{int *} ([\![1]\!] \eta\, \sigma_0)$$

$$= (\qquad \langle \qquad (\qquad))_{int *}\ \iota_{norm} \langle \sigma_0, \iota_{int}\, 1 \rangle$$

$$= (\qquad \langle \qquad (\qquad))\ \langle \sigma_0, 1 \rangle \rightarrow \langle \hat{\sigma}, j \rangle$$

$$= \iota_{norm} \langle \sigma_0, \iota_{int}(1+1) \rangle$$

$$= \iota_{norm} \langle \sigma_0, \iota_{int}\, 2 \rangle$$

$$= \langle \sigma_0, 2 \rangle$$

Volvemos al ej posta

$$(\lambda \langle \hat{\sigma}, z \rangle . \iota_{norm} \langle [\hat{\sigma} \mid t : z], \langle\rangle_v \rangle)_* [\![1 + \text{val } x]\!] \eta_0\, \sigma_0$$

$$= (\lambda \langle \hat{\sigma}, z \rangle . \iota_{norm} \langle [\hat{\sigma} \mid t : z], \langle\rangle_v \rangle)_* \langle \sigma_0, 2 \rangle$$

$$= \iota_{norm} \langle [\sigma_0 \mid t : 2], \langle\rangle_v \rangle$$

$$= \iota_{norm} \langle [\sigma \mid t : 2], \langle\rangle_v \rangle$$

No sé que tan bien está

Creería que ta bien porque intuitivamente deberia dar eso (pisamos lo que apunta x por 2) x es un puntero que apunta a r

r guarda un True

b. η, σ tales que $\eta x = \iota_{ref} r$, $\sigma r = \iota_{bool}$ **True**

η_0 σ_0

$[\![x := 1 + \textbf{val } x]\!] = (\lambda \langle \sigma', r\rangle . (\lambda \langle \hat{\sigma}, z\rangle . \iota_{norm} \langle [\hat{\sigma} | r : z], \langle\rangle_v \rangle)_* [\![1+\text{val } x]\!]\eta_0 \sigma'))_{ref*} ([\![x]\!]\eta_0 \ldots$

$))_{ref*} \langle \sigma_0, \iota_{ref} r\rangle$

$)) \langle \sigma_0, r \rangle$

$=$

$= (\lambda \langle \hat{\sigma}, z\rangle . \iota_{norm} \langle [\hat{\sigma} | r : z], \langle\rangle_v \rangle)_* [\![1+\text{val } x]\!]\eta_0 \sigma_0$

$[\![1+\text{val } x]\!]\eta_0 \sigma_0 = (\lambda \langle \sigma', i\rangle . (\lambda . \langle \hat{\sigma}, j\rangle . \iota_{norm} \langle \hat{\sigma}, \iota_{int}\ i+j \rangle)_{int*} ([\![\text{val } x]\!]\eta_0 \sigma'))_{int*} ([\![1]\!]\eta_0 \ldots$

$= \ldots =$

$= (\lambda . \langle \hat{\sigma}, j\rangle . \iota_{norm} \langle \hat{\sigma}, \iota_{int}\ 1+j \rangle)_{int*}\ [\![\text{val } x]\!]\eta_0 \sigma_0$

$[\![\text{val } x]\!]\eta_0 \sigma_0 = \left(\lambda \langle \sigma', r\rangle . \begin{cases} \iota_{norm} \langle \sigma', \sigma' r\rangle & \text{si } r \in dom(\sigma') \\ \text{error} & \text{sino} \end{cases}\right)_{ref*} [\![x]\!]\eta_0 \sigma_0$

$= \left(\ldots \begin{cases} \ldots \\ \ldots \end{cases} \right)_{ref*} \langle \sigma_0, \iota_{ref}\ r \rangle$

$= \left(\lambda \langle \sigma', r\rangle . \begin{cases} \iota_{norm} \langle \sigma', \sigma' r\rangle & \text{si } r \in dom(\sigma') \\ & \text{sino} \end{cases}\right) \langle \sigma_0, r \rangle$

$= \begin{cases} \iota_{norm} \langle \sigma_0, \iota_{bool}\ \text{True} \rangle & r \in dom(\sigma') \\ \text{error} \end{cases} = \iota_{norm} \langle \sigma_0, \iota_{bool}\ \text{True} \rangle$

Volvemos a $[\![1+\text{val } x]\!]\eta_0 \sigma_0$

$= (\lambda . \langle \hat{\sigma}, j\rangle . \iota_{norm} \langle \hat{\sigma}, \iota_{int}\ 1+j \rangle)_{int*}\ [\![\text{val } x]\!]\eta_0 \sigma_0$

$= (\lambda . \langle \hat{\sigma}, j\rangle . \iota_{norm} \langle \hat{\sigma}, \iota_{int}\ 1+j \rangle)_{int*}\ \iota_{norm} \langle \sigma_0, \iota_{bool}\ \text{True} \rangle$

$= (\qquad))_{int} \langle \sigma_0, \iota_{bool}\ \text{True} \rangle$

$=$ type error

Volvemos al mismo punto

$= (\lambda\langle\hat{\sigma}, z\rangle.\ \iota_{norm}\langle[\hat{\sigma} \mid 1 : z], \langle\rangle_v\rangle)_* [\![1+val\ x]\!] \eta_0\ \sigma_0$

$= (\qquad)_*\ t_{error}$

$\boxed{= t_{error}}$

c. η, σ tales que $\eta x = \iota_{int} 0$

$x := 1 + \mathbf{val}\ x$

Voy directo al paso de evaluar val x

$[\![val\ x]\!] \eta_0 \sigma_0 = \left(\lambda\langle\sigma', t\rangle.\ \begin{cases} \iota_{norm}\langle\sigma', \sigma' t\rangle & \text{si } t \in \text{dom}\ \sigma' \\ \text{err} & \text{c.c.} \end{cases}\right)_{type} [\![x]\!] \eta_0\ \sigma_0$

$= (\sim)_{try}\ \iota_{norm}\langle\sigma_0, \iota_{int}\ 0\rangle$

$= \left(\lambda\langle\sigma', t\rangle.\ \begin{cases} \iota_{norm}\langle\sigma', \sigma' t\rangle & - \\ \text{err} & \sim \end{cases}\right)_{tf}\ \langle\sigma_0, \iota_{int}\ 0\rangle$

typeerror

$= \begin{cases} \cancel{\iota_{norm}\langle\sigma_0, \sigma_0} \\ \sim \end{cases}$

creeria que rebota por typeerror

5. Considere la expresión

$$\mathbf{let}\ y \equiv \mathbf{ref}\ 0\ \mathbf{in}\ (\lambda x.(y := 1; x))(\mathbf{val}\ y).$$

Dar el resultado de evaluar la expresión, y también el resultado de evaluar la expresión contrayendo previamente la (única) β-redex.